

Technology Stack

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3	<i>Used to design a simple, responsive, and user-friendly interface for entering HDI parameters and displaying prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	<i>Flask is a lightweight framework that connects the web interface with the trained machine learning model and processes user requests efficiently.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Pickle (.pkl) Model]	<i>The HDI dataset is stored as a CSV file for training, and the trained Linear Regression model is stored as a .pkl file for prediction.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Flask Server (Can be deployed on Render/Heroku)	<i>The project is developed and tested locally using the Flask development server and can be deployed to cloud platforms in the future.</i>
5	Version Control & CI/CD	Git & GitHub	<i>Used to maintain source code, track changes, and enable easy collaboration and project management.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn</i>	<i>Used for data preprocessing, visualization, machine learning model development, evaluation, and prediction. No external APIs are used in this project.</i>